

SENSOR PLACEMENT VARIABILITY AND DISTRIBUTED MEASUREMENT ARCHITECTURES

General Sensor Placement Philosophy

The wearable longitudinal penile physiology monitoring device may incorporate sensing elements positioned in a variety of spatial arrangements without limitation to any specific configuration. Sensor placement may be optimized for comfort, signal quality, manufacturability, power efficiency, or device miniaturization. The invention expressly contemplates that physiologic information may be obtained from multiple spatial reference locations relative to the wearable structure.

Sensors may be positioned along inner surfaces contacting tissue, outer structural surfaces, embedded internal layers, distributed circumferential arrays, localized sensing islands, or modular sensor regions. Measurement accuracy may arise from distributed sensing rather than single-point acquisition.

Circumferential and Distributed Sensor Arrays

Embodiments may employ distributed arrays of sensors positioned circumferentially around the wearable structure to capture spatial variability in deformation, pressure distribution, thermal gradients, or perfusion characteristics. Distributed arrays may enable localized measurement mapping, asymmetry detection, and improved robustness against localized signal loss.

Array architectures may include evenly spaced sensors, clustered sensing regions, asymmetric placements, rotating reference configurations, or adaptive sensing zones dynamically activated based on detected physiologic conditions.

Inner, Outer, and Embedded Sensor Locations

Sensors may be located on tissue-contacting surfaces to maximize physiologic signal acquisition, while alternative embodiments may position sensors externally or within embedded layers to reduce mechanical stress, improve durability, or protect sensitive components. Embedded sensing layers may utilize flexible printed electronics, stretchable conductors, laminated composites, or encapsulated sensing modules.

Multiple sensing layers may operate concurrently, allowing independent measurement channels to observe physiologic responses from different mechanical reference frames.

Open-Gap and Structural Arm Sensor Placement

In open-gap embodiments, sensors may be positioned along opposing structural arms, across the gap region, or within compliant mechanical members linking structural sections. Placement may support displacement measurement, stiffness inference, magnetic field detection, or mechanical reaction monitoring during physiologic expansion.

Modular and Replaceable Sensor Architectures

Certain embodiments may include modular sensing assemblies permitting addition, removal, or replacement of sensors without altering overall device structure. Modular architectures may support future sensing upgrades, personalized configurations, manufacturing flexibility, or serviceability while preserving core functionality.

The placement strategies described herein are illustrative and non-limiting and encompass present and future distributed measurement architectures capable of acquiring physiologic data through wearable sensing devices.